

ActInf Textbook Group ~ Cohort 5 ~ Session 6

(Chapter 3, Part 1) ~ 10/24/2023

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all right welcome everyone it's October 24th 23 and we're in our first discussion of chapter three for cohort 5 so before we go to any specific question does anyone just want to give any first thoughts or Reflections on three just where it is in the textbook or um what's stuck with them or what they remember from the journey Oli want to give any first thoughts or anyone else yeah sure uh well chapter 3 as opposed to chapter 2's um kind of bottomup approach or the low road uh takes the more daunting or uh let's put it this way the more adventurous route of uh the high road uh which is the one that's been increasingly uh developed in uh the recent years uh so it begins with the notion of what it is it or let me put it this way if we suppose that quote unquote things exist in an environment or in the World At Large what characteristics uh should they possess to consider them as ontological things in other words uh what character characteristics should they possess if we want to uh kind of re-evaluate or remeasure them uh at a specific time scale uh and uh consider them as cons consistently the same thing so the idea is if we put a kind of statistical boundary between the internal states of that object or the thing and its uh external States or its environment then it would be possible to uh to underwrite the Dynamics of um of the of the internal states of the system uh at face of the changing environment given that Mark of blanket or this or the statistical boundary so in other words the internal states of the system um does not change directly um in a response to the changing environment but the relationship between those two states are kind of they kind of vicariously uh influence uh the external States vicariously influence the uh internal state so they don't have direct access but only through the blanket States or Mark of blanket so that's basically the idea for the a self-organized object um and uh it's it's uh one of the main main properties for any living thing or uh rather an idea for uh how life Works um I mean by life here I mean in a very broad term and not just necessarily biological life but it also encompasses any kind of artificial life as well such as xenobots is developed by Michael Levin's lab or any other uh organoids or um similar uh entities that's been increasingly uh appearing in uh many various

disciplines and research areas thank you that's a great summary Kristoff then anyone else who raise their hands um I just wanted to ask because the the concept of um Mark Blanket is sort of a separation between the internal and external environment it always makes me think is that is isn't it really kind of talking about uh temporal SE separation that um ultimately what what really says that we have states that we can identify that are not directly influenced by uh by the environment but but ultimately the system is it is connected so um it's it's it's this this separation like you know eventually breaks down right the signal the signal gets through the barrier and finally influences the internal States there's a lot of pieces there first to last piece yes the signals do get through they do influence so there's causal propagation that can reify the blanket so blankets can exist as interface they do exist as the interface they also only exist over a certain spatial or temporal time scale and then you in the beginning described like time in a maravian sense and that's like a more common usage of the maravian property is through time like the past only influences the future through the present so then the current moment is a markup blanket in time so you could make a markup blanket around any Baye graph you write the Baye graph out you pick the node as internal that's the system of Interest internal states there is a base gra um there is a mark on blanket so that could be um through time you could have one variable through time and talk about maravian property through time it could be spatially embedded and then it could refer to spatial temporal enclosure but the concept does relatively little abstract philosophical work because by the time you define the basian graph it's basically trivial or deflationary where the blankets are and yet it has attracted some of the most heated philosophical discussion yeah go ahead rore so I want to try to connect the formal definition of the Markov blanket as we're using it in this context with what we humans seem to have and maybe probably other animals as well uh which is some intuitive notion of what it is uh to be a thing uh because we look at the at our environment and maybe some of it corresponds to to these partitions that can be defined as Mark blankets uh but we maybe see an animal we see some system and we identify this collection of entities or properties this corresponds to one thing um and of course we can look at it more closely and start to see more individual things inside of that that Collective but well we do have some uh notion of of thingness that's quite uh very intuitive and uh to follow up on that is uh I want to understand what kinds of systems uh we can reasonably describe uh through the idea of a mark of blanket to give an example uh I've heard CL fiston talk about uh setting a a drop of ink inside um maybe like some water a cup of water and the Dynamics of that um ink drop not being maybe not being describable or being describable uh with a mark of blanket uh perhaps the fact that

the ink disperses and doesn't uh maintain it um its coherence is what he means is what is trying to to point out like that means uh this isn't showing that property of thingness and of uh remeasuring that we we we see when we think about a thing um so yeah I want to get a sense for what's what uh kinds of systems whether they're biological or not uh are we is it reasonable to describe using that this the idea of a markable blanket yeah does anyone who have a hand raised just want to speak directly to it otherwise just keep it up and then we'll get to another topic but first just what can you model appropriately with a Markov blanket if anyone just wants to go for it my understanding is that any it's basically anything right because if it didn't have that separation between self and environment there wouldn't be a distinction between environment and the thing itself one uh the drop the example of the drop of ink is one that I actually really like because it shows that a drop of ink will dissipate inevitably but if fit with a living drop of ink just as a single cell yes it would be an constant state of dissipation but also the cell is able to extract uh quote unquote energy from the surroundings right by breaking down some molecules in the surroundings and using that energy to hold itself together and also to use the byproducts to make more components of itself so it kind of counters this dispersion over time um imagining that process to me is you know I'm a biologist and so on but it's still very very hard one example I really like is this bottle I have here oh my background is a little blurred but if I press this button here in this in this shaker I make a little Vortex right and uh it's kind of kind of cool because like it's making something out of nothing it's like you have just environment and you have an input of energy that set that makes a a low lower end trophy system right that demarcate itself from its environment which is made of the same thing as as this thing and all you need as a battery is dissipating a gradient right if you don't have the gradient you have no fluid Vortex and one thing on the that know they that that I don't know if somebody has an answer is uh what is a mark of does this thing have have a markco Blanca just the kind of flame is and if it does well it it has to right because it's a thing separated from other things what is the can we classify different kinds of Micco blanks you know like this is this is very transient it's very thin it's so but that membrane the membrane of a cell is a more much more complicated blanket um many many points just to very briefly note them because it's again it's always more time in between our discussions like yes it is about the cell and the ink here's the one of the earliest 2006 examples there's a snowflake so snowflakes are getting whipped around like your Vortex external energy is going through the so what can you say about a snowflake well it's frozen but what could you say if there were some snowflakes that also could preclude melting so that's why it has to start with a thing and a lot

falls out from you starting with a persistent thing or a remeasured thing does a thing have a markup blanket this is the Eternal realist instrumentalist question the realist language is yes it has a Markov blanket cells do active inference this organism has a Precision estimate the instrumentalist position is you can model it with a Markov blanket it's as if it has a Precision variable and so whether we say that it's about what the territory is doing about what it has or is versus what we could do classifying different kinds yes absolutely transient watering blankets these are the kinds of things that many people are researching about because when you actually do have a system of Interest these kinds of questions are just immediately the things that you need to ask um Jesse uh or or yeah go ahead David sorry okay Jess may I just interject for a moment to add another angle sure this yeah okay so um another perspective for viewing a Markov blanket um uh I mean whether or not it is ubiquitous or or just uh it only applies to specific things is that uh okay so theoretically it is possible to draw uh draw some statistical boundary between anything that is sufficiently separated from its environment but uh for the things that follow or that have a kind of linear causal structure uh to their Dynamics or to their interactions with the with the environment this Markov blanket doesn't say anything specifically interesting about the Dynamics so uh the significance of Markov blanket uh is becomes more apparent uh when we we're dealing with the systems with nonlinear causal structures to their interactions with the environment so the more uh nonlinear their Dynamics the more uh interesting things Markov blanket can say about uh the uh the the interaction of that system and it can uh somehow result in non-trivial um non-trivial elements of um describing of the description of the system so uh as Dalton St Vel has shown in one of his paper week uh blankets uh it's not uh I mean the question of Markov blanket it's not a question of whether or not but it's a question of degree so we can have uh a Continuum of uh blanketed uh uh if you will of different uh different degrees of strength for that Markov blanket so again for linear systems we have very um trivial kind of Markov blankets and as the system get get more complicated or more complex and uh with higher dimensional um structure a Markov blanket gets become increasingly non-trivial thank you oi um Jesse and then and then Susan yeah that's really interesting what you're saying Ali because I mean the way that I've been viewing the Markoff blanket is I know that there is I guess a temptation to extrapolate like a broader meaning about existence and thingness but I'm just kind of looking at it as a statistical tool to you know delimit inputs and right it's like it's a way of organizing inputs and defining external States and internal States because in the things as they are the the number of variables of inputs is really difficult to you know quantify in that that broad causal chain and so that's

maybe where Ali that what you're saying about the nonlinearity is is really interesting um but in model making where we are like investigating specific input output relationships the Markoff blanket seems to be very useful for us to be able to you know create that type of basian graph and organize that information great yeah and also like blanket only exists as part of the particular partition in relation to the rest that are being described so it's not like a blanket drawing exercise it's like a system modeling exercise and then taking different system of interest um as your internal State you just by definition get a marov blanket so hence it does little philosophical work by itself and yet has attracted some of the most attention so so um so aren't blankets um at isn't there blanket versus blanket in other words there's different scales and when you say partition that you know that kind of maybe are you when you say partition are you talking about um Markov blankets at different levels and and to you to kind of tie on to what Jesse was saying art Markoff blankets context specific okay a lot there let's think about a few more tangible pieces first here's the definition so we're talking about the relationships of variables not about the relationships of things in the world so any discussion of what a markup blanket is slash does as the technical concept it's not going to be supported or rejected by any feature of the world it's about the variables being related and then the partition is just the definition of what the markco blanket is doing which is a set of nodes one or more that are intervening between two other sets that comes into play when modeling how variables are connected so if we made a model of uh temperature in the room and there was just one thermometer carrying forward through time we might have one model where every time Point can influence every other time point but then for long time series that's going to get out of control so a common modeling assumption is like well make it a Markov chain a one Markov chain where only the last time step influences the next time step so that was what was mentioned earlier like that's a pure Markov blanket in time is that a claim that there isn't like time effects in the real world well not specifically but then you could at least use that and say actually we're finding that the one marov chain model Capt pretty much all the empirical phenomena so still that's not saying that there aren't some time effects in some setting but just saying that it's totally sufficient for that case so so one more question that for me ties it up is that when you say Markoff Chain versus the model I'm kind of relating a markof chain to a map yeah I think you can even go further the whole thing is a map we're only talking about cognitive cartography only map and then the marov chain is this Motif where you have a set of chained States hidden States through time with a markof transition Matrix B that's the one marov property is that is that horizontal chain and so like you could have a probability Matrix

that is like on the diagonal it has things staying the same Sunny to Sunny 0.9 and then cloudy to cloudy 0.9 and then the off diagonals is like the transition between them so those are some simple statistical models of dynamic systems and a Markov chain is when they're just chained forward in time the Markov blanket could be considered from a static perspective again that's not to say like organisms are or aren't Dynamic just about the Markov blanket you could have a causal Network that is static uh it's how Bert in his work in 55.0 today introduced the Bas graph concept was with a static causal chain just kind of contextual factors interacting like a structural equation model but in a basian setting and then you can imagine composing a Bas graph that has um a spatial and a temporal component or um a temporal and a perceptual component or a temporal and a perceptual and an active component and then when we have a base graph in time with something like perception and action then that is where we get the partially observable mdp but like Markov decision process was a term before and outside of this kind of Markov boundary as organism boundary discussion Markov decision process is just referring to the Markov chain and making decisions on a Markov chain yeah it's an interesting top and it's the heart of the high road as Ali mentioned because whether we're from the outside talking about remeasuring something that's like so then there's things that are static compared to how we're remeasuring them there's things that are changing faster than we can remeasure them so then they're not being detected as the same thing like gas molecules in room but then there's some spot where there is like a temporal thickness in modeling and then that could be seen from either the remeasurement the experimental position which comes back in chapter nine looking like this experimenter observing something or to persist in time that can be framed as a remeasurement of homeostatic attractors rather than for example the maximization of a reward or utility function it's like well now the zeroth level task is to remeasure in the homeostatic range not to be rewarded by values in the homeostatic range and then maximize reward but to directly Target the statistical measurement any other topics or or quote to a section that somebody wants to like go to yeah yeah just this whole relationship between um value and belief you know the relationship between the value function and and um know the the function of Prior beliefs and surprise minimization I thought was a really interesting uh jump that I'd love to talk more about or to have some people's thoughts on which page section or anyone can just give it a thought section 3.5 page 53 yeah very related to the previous discussion replaces the notion of value with the notion of Prior belief so we can contrast the sort of imperative to be value and so so the the the um the base assertion to be value exists and then the imperative to be that it has to be

optimized or maximized alternatively a sort of foundational assertion can be the thing exists which might be kind of a neat assertion because we might only be interested in those systems that do fulfill that and then the imperative would be the kind of more boring and first level be that thing whatever it is be as it is as um and instead of setting up even anticipatory so so that that seems to to more closely apply to like okay the rock is just sitting there the bird is just flying okay things just kind of do their own thing but then that's the variational free energy so then the next question is well then how do you actually get snowflakes that fly and resist melting cells that don't just dissipate like an ink drop how do you actually get something to both be what it is and have a preference or realize its preference or have agency in realizing its preferences and that's where the expected free energy comes in again making no assertions about the mechanism that any given system actually does to to fly and avoid danger but giving a statistical framework for the bird watcher so that they can say something statistical about the behavior that they're seeing and then the other side of the coin is like the mechanistic and and the and that's the low road H how is the bird or the bat avoiding danger well it's not actually doing matrix multiplication but it's a whole question to actually investigate how and that's the low road for for each system but the high road is kind of the starting point for discussion because without the high road it's like an admission that it's a purely hypothetical space and then even that exists in a sense inks and then anyone else I just wanted to give an example about this um in my work often people act against their goal because the I was going to say value but the the intrinsic reward for proving their unconscious prior belief I am the one who suffers I am the one that everyone hates somehow the organisms values that higher this prior belief of identity I am that thing than actually achieving a goal that they set for themselves so to to me intuitively that makes a lot of sense that swap there because I imagine that has to do with the um deep affect that comes with prision evaluating your prior is good like my old systems work my old priors are valid so there is a high there around good affect I may be suffering I may be losing but I was right all along can't take that away from me yeah lot lot to say there so in the affective inference work they linked the psychological phenomena of affect to Precision estimation so if things were more variable than expected that was negative AFF effect if things were more precise it was positive a effect and if things were as precise as expected as variable as expected it was neutral balance and so if there's an empirical track record of something being a certain way like I've gotten out and gotten \$1 each day I'm the kind of person who gets \$1 and then now there's an affordance that is empirically not been taken then there's these different mechanisms to explore like Precision on policy or Precision

on sensory inputs maybe the three is like surprising and so then it's like less cognitive overh
good or just more natural flow to go back into an empirical previous way of doing it and but
then multiple models are constructed it's not just like well a toy model was made and now this
is like you know people are going to be using this in a clinic it's like multiple models are
iterated in in a lot larger and a slower setting over the years and then more and more is
understood about computational Psychiatry um Kristoff then anyone else um I find it uh
interesting to to think about this uh you know um this effectively functional of functional
beliefs as as a sort of you know this is the This Is The Meta function that we're trying to
optimize and and I often find it that when I talk to people about for instance the uh the
notion of Free Will um I like to say that we don't really have free will because you know if
you consider really your circumstances very carefully you will find out after um you know uh
sufficient analysis that there's usually the correct uh way to act and uh you're only it only feels
like you have this free will because you're acting under uncertainty and you're making a
decision for for for the first time so it it feels like you're doing something because there is
there is this place there is a wiggle room but um you know this this this really nicely fits with
this you know notion of Prior beliefs that sort of uh those prior beliefs make you what you
what you are you are a thing because you have prior beliefs and I like to use the uh the term a
Aesthetics um because that really determines you know um uh kind of the kind of values
you're going to pursu is the things that the things that you care about the uh the the the things
you find beautiful the things the way the way you want to be that's kind of Aesthetics but the
the the process um of you know so you can you can in some in some way you can choose that
but the process of how you're going going to get to your preferred States that's predetermined
by the by the by the physics of reality right so that's uh and and that's where the kind of like
the this disension between having having free will having the the wiggled room having
Aesthetics that you can choose having the prer beliefs that you can that that you kind of have
to well you can't you can choose them um if you learn how then you you get to adjust your
Precision parameters right it's just a m matter of practice to get to get the root access to to the
mind to be able to do that um and then and then everything happens is kind of automatically
from that point on because if you're a free energy minimizing agent you always know what to
do next find the the the the juiciest pocket of negative entropy Goble it up and and uh go about
your business yeah I mean saying that a thing is or or even has priors is the definition of
realism certainly a stance that people have asserted but also people can use the same
formalism on that exact same system of Interest or a different system of interest in support of

an instrumentalist claim so the framework can be used to advance a realist claim but at least from my perspective that doesn't make mathematics an assertion about any specific system or how any specific thing is it's literally to the exclusion of that that we have a general framework Susan um so I I actually put about four questions in the in the notes I didn't know if you were going to try to get to the questions um in uh shall you look at it from here or should I go to the chapter or core five questions here uh the questions yeah yeah Y and and I guess specifically yeah what what's which one what word to what word to look up to find it affordances okay um and and that yeah kind of actually pulls on both the thank you the question in terms of inks and Kristoff's um discussion in terms of you know are these can these be learned um okay great thank you for adding them so Ali explains in the intro video at the top of the page Markov blanket is a boundary in the state space not necessarily a spatial temporal boundary came up today as well what allows the agent to be statistically separated from from its boundary as observed through its blanket so I take it that the blanket is a lens through which the agent observes environment yeah lot lot of ways that that that could be taken but yeah if we if we kind of associate why with observation and the chain of perception yes the inbound dependencies across the blanket do map to like a lens and then also it's a great um connection as some will probably see that actually in the category Theory like the lens there's probably even a deeper meaning of this too but for so we won't go into it but short answer yes would this lens be defined by their so the outgoing dependencies from the blanket so literally the other side of perception action outbound dependencies the space of the possible actions is defined by affordances but if the hidden state were a person then what their affordances would be you know the observations of the other person so these are not tags on the world it's not like this object is an observation this object is an action it's like in this map with this as my internal State then this dependency can be read as perception or as action affordances according to Gibson must be learned yeah I mean there's really two things here there's kind of like Gibson Aly like what did this specific person think about this word which I'm always all for but the short answer is aordance is being used in a slightly different way in active inference not to refer to the direct agentic perception of an action capacity but a little bit more from a modeler's perspective as just simply the action capacities whether or not they're perceived without making any statement on whether they have to be learned or not we can model a pendulum with active inference so obviously the affordance concept doesn't need learning you can just model it without learning does the distinction between active learning and active inference have to do with these affordances so yeah this was a distinction that that one paper in the

Discord mentioned short answer here is you could have two agents with the same set of affordances. Move up down left right and then you could set up the parameters differently so one of them sought to confirm its expectations about sensory another one sought to reduce its uncertainty maximally about world States so those are all the parametric fine-tunings but it doesn't require a difference in affordances but it might require difference in how the affordances are being taken that's how you would know that the behaviors were different at all you could also make a model where there were different affordances so that's for situational modeling and then the question becomes very obvious are internal State parameters referring to the edge of the Markov blanket State space not exactly the internal State parameters are everything about the the system of interest that isn't the interface so it's like the shell of the egg is the Markov blanket and then the inside is the internal States because like when you take a photo you're just going to see the shell so that's just the outer interface with perception action across the blanket and then internal State parameters are everything about that thing that isn't the shell and then together the thing is internal plus blanket because that's what moves around relative to the does anyone else have any like kind of other other takes on any of these questions yeah Oli please uh yeah so uh there's another uh rather simplified um representation of Markov blanket with regards to uh the probabilistic dependencies of the different states uh in uh the physics of sentience U presentation by K. Friston uh so if you search for uh this topic in YouTube there are several different presentations but around uh the the time four minutes or something you can see a clear uh representation of Markov blanket which shows how exactly external States internal States sensory States and active States relate to each other according to their probabilistic uh dependencies so I think that can be quite helpful cool also yeah a lot of other lot of other pieces these are all these are all great great questions and it's one reason why there's such an emphasis on making the generative models because there is such a difference between the abstract relationships purely within the actim fonty just the sort of conceptual relationships and then when mapping it to a system of Interest it's kind of like a 90 degree rotation with what you're connecting up connecting ACM terms to specific things and then when this is all accounted for that's all accounted for then that's the generative model but the textbook is describing like the pure relationships with a um stream of examples and boxes and kind of just different um pieces different discussions another interesting thing is like the high roads describing the imperative but the low road is there's might be multiple ways to realize it so just again to separate out how it's being done from why it's being done um Verena thank you go for it hi I have a question and it might be um uh connecting to Chris of the uh question

of um free will there is a sentence in chapter three and I don't know where I'm so sorry but um what it says is that there is this um um when you maximize actually the uh free energy where you keep your options open until you have and I guess does it mean until I have enough evidence that I uh can state the belief and would it be a demonstration of Free Will that's my question that's a great great question it's kind of cool it comes up twice in the low road so kind of as a previously identified pattern of Bayesian Statistics just having nothing to do with um modeling cybernetics and then here the maximum ENT y principle of Janes the imperative to explain things as accurately as possible but also keep options open Oli do you want to give a thought on this uh so yeah I think uh the question of Free Will related to active inference you see because um as far as I understand it um active inference is a kind of framework uh that puts agency at its Center Stage right uh so uh the minimal definition of agency for for us to uh endow or or confer agency to any system uh can be uh put it put in its most minimal sense in this way so if uh if there is more than one way to respond to a stimulus uh and and uh the other condition would be uh if the the system or the agent has a way to choose among uh those responses then uh we can probably say uh it has some minimal sense of agency so in this sense I believe active inference models or um or the agents that's modeled with um markup blankets in active in in active inference Frameworks uh they have indeed this kind of minimal agency to them but if we go a little bit deeper than that and U consider uh the deeper generative models and deeper hierarchies then uh we would come across agents with beliefs about their beliefs H and we can of course continue that ad infan theoretically so beliefs about beliefs about beliefs and so on uh and uh the more deeper the generative model gets uh the more sentient or let's say uh probably more conscious the agent will become but in any case even for the most simple case of uh active particles we can um describe agency with regards to these minimal definitions so I believe uh active inference and Bayesian mechanics put uh agency or rather Free Will um as some of its core tenants tenants for uh describing those systems so it is written to the core of active inference uh this conception of agencies and uh it's something that we cannot take for granted uh even if that kind of agency is not uh so real agency but just a kind of apparent agency or in other words a kind of agency that appears as if the system have has so um yeah I I'm not sure how uh uh how IT addresses the question of whether there is uh whether or not there is kind of free will for these kinds of Agents but um I think uh it can be uh seen as some kind of U minimal framework to um address these questions as well great I'll offer another position which is the maximum entropy principle also applies to gas molecules in the room and the temperature so I think even in this discussion we've heard some um

positive and negative attributions of of free will an otive inference but then the question that that we can still circle around is well what's the relationship between the technical and analytical maximum entropy principle connected to all this other stuff for example the constrained maximum entropy principle being dual to the F more recently being found out but how does any of those principles relate to how some body mobilizes an account of a certain biological system for example that's even a different kind of question Susan yeah well along those lines I was just reading something about metacognition and um and some studies done um that uh that relate um the difference it made when um when students were actually taught how to think about their beliefs differently in the um and so I think that that for me that kind of brings in the the belief updating and the learning um you know as part of agency because even though I have different options I may you know my I'm assuming that you know the Markoff blanket can also retract depending on my state um and you know that's going of um hollow out opportunities um I can't see my affordances even though they're there would that be would that be um you a am I pulling those threads together correctly I guess yeah that's great thank you just so we can hear from everybody in the last minutes um Brena then Kristoff then Jesse um and I have another uh question it's page 58 um the first sentence furthermore active inference is largely coherent with idomoo Theory uh which states that action starts with an imaginative process uh has to been some work done what this uh imaginative process is or um or what what's part of the process and how it is um yeah is there any any research group do you know anyone or does anyone know about um this kind of imaginative process which triggers the action that's a great question I think you'd find that in essentially all prospective ACM generative models you could interpret it coherently with idoor like if you chose to associate um policy selection with ideas and then the policy that gets selected would would have Arisen from a imaginative compositional process that triggers action there may also be people working more on the um psychological side but not I mean you can look it up or or post in Discord or continue to explore but but I think that it's a cool question topic because uh um I mean imaginative process is a process viewed from a human not really from maybe a cell or or something I guess I don't know that's uh that's why I was like a bit triggered by by that uh expression so um I wasn't sure about what it was addressed here but thank you for for the clarity I will yeah it's like we we we come to it from so many different way well what if a computer program makes new compositions is that imaginative not that doesn't need to be the claim that um a a human process is the same as a computer process but if they can be modeled with the same formalism then from an instrumentalist perspective it's just like saying well we

could measure a squirrel and a bird with the same meter stick that that makes limited claims about different animals and how similar and different they are Jesse and then anyone else yeah I I like the really thinking about that relationship between variational free energy and maximum entropy uh sort of in the risk management framework in a way it's it's it's it's like engaging in this you know it isn't just a matter of like prediction error minimization but also risk managing predictions um in a certain respect so that that that topic just made me think like that yeah it would be a risk to over or underfit if you underfit the distribution you'd be leaving information out you wouldn't have as much awareness as you could if you overfit it would be fallacious in a different way so you'd want to be at the right distribution given what you knew yeah it's an interesting framework totally agreed um Resa then Kristoff um hi uh so uh going back to um something Jesse raised about section 3.5 on the active inference side where it begins by describing uh how Behavior results in reinforcement learning versus uh active inference I was just thinking from a practical perspective has there do you know of any examples where um the simple reinforcement uh learning examples have also been implemented in code uh to demonstrate how the active inference approach to that would be like the card po card pole example or some some something um yeah Cod tangible cool the code page you'll find several dozen um applications and especially the work of um Alex shance Norse AED Connor Hines others uh Baron millage those tend to directly tackle and juxtapose non-active inference machine learning and and we've had many live streams on the mountain car the multi-arm Bandit several other kind of classical settings that make the comparison pretty um clear Kristoff and then anyone else with a last word uh yeah I just want to say about this you know keeping keeping your your options open it always reminds me of this um fantastic uh thing that the inestimable Terence McKenna said in one of his Raves that you know he he was asking this question like why us why is the mushroom revealing the secrets of the universe to us and he said like well uh because I I don't believe in anything I I entertain ideas and I I try to apply it definitely in my life is that you know whenever I think I'm convinced that I I understand something that that that this this is true and that is not true um I look back to that and also to the teaching of of the Buddha and and think to myself like no the answers are never really that simple that's like the question of free will simply might might not have a binary answer it's just simp as simple as that it's it's it's complicated it's a process and we may not be able to determine one way or another because it simply might not be um uh determinable that way the fact that the syntax permits us to make a statement doesn't mean it's a valid statement to make um so that's uh keeping your options open yeah that's a

that's that's an important thing to meditate on um you know every day what do I believe
nothing I gather evidence for um for the existence of ideas lot of different perspectives to take
great times well thank you everybody we will come back next week in the earlier time for oh
will be Halloween chapter 3 a spooky Adventure okay um costumes are [Laughter] optional
thank you everybody see you next time thank you everyone bye bye bye thanks everyone bye